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80 ECHO MEASUREMENTS FROM FLUTTERING INSECTS SUGGEST THE ACOUSTIC BASIS FOR COMPLEX TARGET PERCEPTION IN THE FM BAT, EPTESICUS FUSCUS. *C.F. Moss[1], J.A. Simmons[2], J.D. Park[2], [1]Dept. of Psychology, Harvard Univ., Cambridge, MA; [2]Dept. of Psychology and Section of Neurobiology, Brown Univ., Providence, RI

The echolocating FM bat, Eptesicus fuscus, perceives target distance from the delay of reflected echoes and target depth structure from the spectrum of reflected echoes (Simmons, JASA, 54: 157-173, 1973; Simmons et al., Science, 186: 1130-1132, 1974). A sonar target with multiple reflecting elements creates a spectral interference pattern in the composite echo, with frequency regions of cancellation and reinforcement that depend upon the spatial separation of the target surfaces. The bat's auditory system initially encodes the frequency regions of cancellation and reinforcement along the axis of the spectrum; however, the bat's auditory system converts the spectral representation of a target's 3-D structure into the underlying delay (or range) separations between reflecting surfaces (Simmons, Moss and Ferragamo, J. Comp. Physiol. A, 166: 449-470, 1990). The transformation of echo spectrum into discrete estimates of range separation between reflecting elements provides the basis for the perception of target depth structure in Eptesicus. In this study, we attempt to integrate these recent behavioral data on the perception of depth-structured targets in FM bats with measurements of echoes from fluttering insects. Our goal is to develop models of acoustic image represesentation of natural and moving sonar targets in FM bats.

Computer-synthesized FM sweeps (100-20 kHz) were amplified (Krone-Hite) and broadcast at a tethered, fluttering wax moth through a custom-made ultrasound speaker. Echoes were picked up with an ultrasound microphone (Ultrasound Advice), amplified (PAR), filtered (Rockland) and recorded at 60 ips (Racal). Echoes were also digitized at a rate of 250 kHz (RC-Electronics). The echoes were analyzed off-line using a real-time sonograph (Kay 5501) and signal processing software (ILS). The echo measurements from fluttering moths show systematic changes in echo amplitude and spectrum that could provide reliable information about target shape and wingbeat rate. These findings are presented in the context of our current understanding of acoustic information processing in FM bats.

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81 THE TEMPORAL RESOLVING POWER OF A UNIQUE VERTEBRATE AUDITORY SYSTEM. *S.E. Allan, A.M. Simmons, Dept. of Psychology, Brown Univ., Providence, RI

The two main vocalizations in the male green treefrog's (Hyla cinerea) repertoire have similar spectral components but differ in the temporal property of rate of amplitude modulation (AM). These acoustic differences carry biologically-relevant information, which might be reflected in the male's behavioral sensitivity to rates of AM. We are currently undertaking a series of experiments to investigate the temporal resolving power of the male green treefrog's auditory system using field playback, laboratory evoked calling, psychophysical, and physiological techniques. Different rates and depths of AM are superimposed on digitally-generated complex sounds whose frequency components match spectral peaks in species-specific signals (AM calls) and on band-passed noise (AM noise). These various stimuli are presented at intensity levels spanning those at which males communicate in their natural environments. Field playback experiments show that males discriminate between different AM rates primarily in terms of the timing of their vocal responses with respect to the stimuli. Males tend to call more in response to AM rates matching those of species-specific signals, for both AM calls and AM noise. In the laboratory, males respond most consistently, with shortest latencies and more calling bouts, to unmodulated (synthetic advertisement) calls than to calls modulated at 50 Hz (synthetic aggressive calls). At high intensity levels, both in the field and in the laboratory, calling is rather unselective, but definite audiograms can be measured at lower intensities. Psychophysical results suggest that the male is capable of fine-grain temporal resolution. The shape of the temporal modulation transfer function, measured both psychophysically and by midbrain evoked potentials, reflects the coding of acoustic features in species-specific vocalizations.

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82 HEARING IN SUBTERRANEAN MAMMALS: NAKED MOLE RATS HETEROCEPHALUS GLABER. *R.S. Heffner, H.E. Heffner, Dept. of Psychology, Univ. of Toledo, Toledo, OH

Naked mole rats are small (20-40 gm) rodents with uncertain phyletic affinities. They are of special interest for understanding the effects of underground living on hearing because their specializations for a fossorial lifestyle are extreme: loss of hair, minute eyes, no external pinnae, and poikilothermia. Unlike the other fossorial species whose hearing is known, naked mole rats are not solitary but live in large colonies (with a single breeding queen and numerous sterile workers) and they vocalize frequently.

The complete hearing range and sensitivity were behaviorally determined for 4 naked mole rats using a conditioned avoidance procedure with a food puree reward. The animals responded to frequencies from 8 Hz to 16 kHz with a hearing range at 60 dB from 65 Hz to 12.9 kHz. The animals were remarkably insensitive throughout their hearing range with their most sensitive hearing being only 35 dB at 4 kHz. Alternate means of detecting low frequencies such as substrate vibration or air movements affecting vibrissae were ruled out.

The most common vocalizations of naked mole rats are brief "chirps" (approx. 200 ms) beginning near 4 kHz and falling to approximately 2 kHz. When disturbed, they "bark" with a fundamental near 750 Hz. These vocalizations are in the frequency range of their best sensitivity.

The audiograms of the three fossorial species which have been tested behaviorally (naked mole rats, blind mole rats, pocket gophers) share two features: (1) very restricted high-frequency hearing limits (comparable to non-mammals) that deviate significantly from the strong correlation between interaural distance and high-frequency hearing found among other mammals, and (2) poor sensitivity throughout the hearing range. However, the three species differ in their low-frequency hearing with both naked and blind mole rats possessing sensitivity to low frequencies but pocket gophers having limited low-frequency sensitivity similar to that of high-frequency rodents such as Norway rats.

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83 HEARING IN LARGE RODENTS: GROUNDHOGS MARMOTA MONAX. *J. Conesa, R.S. Heffner, H.E. Heffner, Dept. of Psychology, Univ. of Toledo, Toledo, OH

Groundhogs are of interest both for their large size and because they belong to the group of rodents which nest underground thus providing additional information on the role of underground living in the evolution of hearing. The complete hearing range and sensitivity of 4 ground hogs were behaviorally determined using a conditioned avoidance procedure with a water reward. The animals responded to frequencies from 32 Hz to 32 kHz with a 60-dB hearing range from 40 Hz to 27.5 kHz. Their best hearing was at 2 and 4 kHz.

The low-frequency sensitivity of ground hogs places them in the group of rodents with very good low-frequency hearing--that is, a 60-dB hearing limit below 100 Hz. All of these, with the possible exception of chinchillas, share its lifestyle of nesting underground. Even though it is possible that such a lifestyle may promote the development of low-frequency hearing in small mammals, it does not seem to be sufficient since at least two species, the pocket gopher and the grasshopper mouse also live underground but are not sensitive to low frequencies.

The second significant feature of grounghogs is that they are the largest rodent (3.5 - 6 kg) whose hearing has been determined. With 20 species of rodents now available we can examine the relationship between interaural distance and high-frequency hearing within a single phyletic group. This correlation among rodents is -.74, which is very similar to the correlation of -.81 among all mammals. The relationship extends to the large groundhog as well as to small mice and applies equally to surface dwellers and to those which, like the ground hog, nest below ground but forage on the surface. However, the three exclusively fossorial species deviate significantly from the correlation due to their extremely limited high-frequency hearing despite their small size.

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